MANCHESTER 1824
The University of Manchester

Comparing Arm Tracking Teleoperation to Conventional Joystick Control of a Laterally Mounted Robot Arm

A dissertation submitted to The University of Manchester for the degree of

Bachelor of Engineering in Electronic Engineering

in the Faculty of Science and Engineering

Year of submission

2025

Student ID

11006231

School of Engineering

Contents

Contents 2

Abstract 4

Declaration of originality 5

Copyright statement 6

Acknowledgments 7

1 Introduction 8

1.1 Background and motivation 8

1.2 Aims and objectives 9

2 Examination of existing human to robot arm teleoperation methods 9

2.1 Why choose teleoperation over conventional input methods 9

2.2 Requirements of a suitable tracking system 10

2.3 Survey of existing tracking technologies 11

3 Comparing two viable computer vision based methods of robot arm teleoperation 13

3.1 Comparison of both computer vision methods 13

3.2 Results of joint angle tracking methods 16

3.3 Final Tracking Method 17

4 Arm tracking vs conventional control experiments 18

4.1 Description of experiment to compare conventional vs human arm tracking 18

4.2 Scenario 1: Random position matching task 18

4.3 Scenario 2: Simple environment pick and place task 19

4.4 Scenario 3: Obstacle-rich environment pick and place task 20

5 Conclusions and future work 22

5.1 Conclusions 22

5.2 Future work 22

Appendices 23

A Project outline 23

A.1 Overview and motivations 23

A.2 Deliverables 23

B Risk assessment **25**

C Code **26**

D DH Parameters **26**

E ANOVA Figures **26**

Word count: 1234

Abstract

Abstract goes here.

Declaration of originality

I hereby confirm that no portion of the work referred to in the thesis has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

Copyright statement

i The author of this thesis (including any appendices and/or schedules to this thesis) owns certain copyright or related rights in it (the "Copyright") and s/he has given The University of Manchester certain rights to use such Copyright, including for administrative purposes.

ii Copies of this thesis, either in full or in extracts and whether in hard or electronic copy, may be made *only* in accordance with the Copyright, Designs and Patents Act 1988 (as amended) and regulations issued under it or, where appropriate, in accordance with licensing agreements which the University has from time to time. This page must form part of any such copies made.

iii The ownership of certain Copyright, patents, designs, trademarks and other intellectual property (the "Intellectual Property") and any reproductions of copyright works in the thesis, for example graphs and tables ("Reproductions"), which may be described in this thesis, may not be owned by the author and may be owned by third parties. Such Intellectual Property and Reproductions cannot and must not be made available for use without the prior written permission of the owner(s) of the relevant Intellectual Property and/or Reproductions.

iv Further information on the conditions under which disclosure, publication and commercialisation of this thesis, the Copyright and any Intellectual Property and/or Reproductions described in it may take place is available in the University IP Policy (see `http://documents.manchester.ac.uk/DocuInfo.aspx?DocID=24420`), in any relevant Thesis restriction declarations deposited in the University Library, The University Library's regulations (see `http://www.library.manchester.ac.uk/about/regulations/`) and in The University's policy on Presentation of Theses.

Acknowledgments

Acknowledgments go here.

1 Introduction

The goal of this project is to create a system for controlling and teleoperating a robot arm with a human arm. In order to maintain simplicity in the project scope, the robot arm is assumed to be shoulder-mounted and has the seven degrees of freedom (DoF) **ref:arm_dofs**, matching that of a healthy human arm. An example of the humanoid robot arm (the YuMi robot) is shown in figure 1.

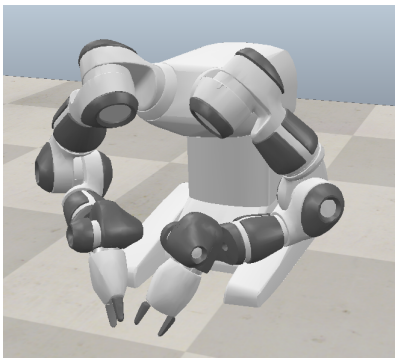


Fig. 1. YuMi Robot Arm To Be Used In Experiment

1.1 Background and motivation

The motivation for this project comes from articles within the books: **ref:Zhou_Yang_Wang_Dong_2025** **ref:Zhou_Yang_Wang_Dong_2025** and **ref:Lyu_Yang_Yang_2025** **ref:Lyu_Yang_Yang_2025**. Both of these articles use an inertial measurment unit (IMU) to gather data, but only maps the wrist position to the robot's end effector.

Redundancy in the degrees of freedom of the arm means that a valid end effector position can have multiple positions of the previous links **ref:nullspace**. Therefore this ambiguity could lead to collisions with obstacles in the environment, and lack of safety for people within the envelope of the robot arm.

Some specific use cases will require absolute control of every link of the robot arm - for example underwater sample collection. Whilst **ref:Jin_Ji_Yan_2023** describes a potential autonomous solution **ref:Jin_Ji_Yan_2023**, this method is comparatively slow to the prospect of remote teleoperation. Consequently, the complexity increase for the case where nothing is stationary, a more realistic use case, would mean that this method would be inefficient compared to the intuition of a trained operator.

Additionally, **ref:vranalysis** shows that for tasks involving teleoperation of a robot for pick and place

tasks, virtual reality style mapping of head movements with arm to end effector mapping is more efficient and less error prone than a typical joystick **ref:vranalysis**. Hence, exploration of the performance characteristics between conventional control methods and arm mapped teleoperation is undertaken.

1.2 Aims and objectives

The overall aim is to compare the performance of a system for mapping the joint angles of a human arm to a shoulder-mounted robot arm to conventional control methods. This will include:

- Identifying and comparing existing technologies, including IMU based solutions discussed earlier, attitude and heading reference systems **ref:Mazomenos_Biswas_Cranny_Rajan_Maharatna_Achner_** computer vision based systems **ref:Phuong_Cong_2024**, and flex sensor based systems **ref:Nalam_Maniva**
- Developing a version of the tracking system suitable for controlling a simulation of the YuMi robot arm.
- Devising experiments appropriate to comparing conventional methods of control to arm tracking teleoperation methods.
- Analysing the difference, if any, in performance between either method of control in a variety of environments.

2 Examination of existing human to robot arm teleoperation methods

This section explores which arm tracking methods are the most suitable for this project. Quantitative and qualitative data from secondary research is used to create a justified specification for the final solution. Finally, a Pugh matrix will be applied to fairly score and select an appropriate method to further develop.

2.1 Why choose teleoperation over conventional input methods

The measurement of human arm joint angles has been an interest of academia for several decades: with one of the first examples being demonstrated by **ref:videobasedumanmotioncapture**. In in-

dustrial settings, full body tracking has primarily been used in the film industry to accurately measure an actor's performance **ref:mocapwhy** and map them onto computer generated characters **ref:whatmocap**.

This approach could also be applied to the control of robotics: have a skilled operator remotely control industrial robots where traditional control methods (e.g. joysticks) are unsuitable. Research has shown that this approach could lead to 1.8 times faster task completion time and with a quarter of the errors (1.54% compared to 8.65%) in the best cases **ref:vranalysis**.

Another way to increase the performance of a task with teleoperation is mapping the head movements of an operator to the camera of the robot performing the task **ref:teleopanalysis**. However, this study was based primarily on the control of wheeled robots; manual joystick controls exceeded the performance of teleoperation in this specific task.

These two cases suggests that whether teleoperation is more effective than traditional joysticks is strongly dependent on the task. Overall, in the case of robot arm control, there is potential for performance increases via human arm teleoperation **ref:Zhou_Yang_Wang_Dong_2025**, strengthening the justification for this project.

2.2 Requirements of a suitable tracking system

In order to compare the final system to existing systems and research, desired characteristics must be identified.

Accuracy requirements - Be able to measure joint rotations with a minimum resolution of $3°$.

Experienced surgeons have been measured to have $5°$ of vertical movement range on all joints when asked to hold their hand on a target, and close to $3°$ range for horizontal movement **ref:movementrange**. Therefore any measurement lower than this resolution can be characterised as 'noise' from human inaccuracy, and can be smoothed out.

Latency requirements - Have a maximum system latency of 300 ms.

Teleoperation task performance generally degrades around the two key figures of 600 ms and 1500 ms **ref:teleoplatency**. Thus, the Nyquist criterion equivalent of this threshold is used as a well informed estimate of how much latency can be present in the system, leaving a final requirement of 300 ms.

Resource requirements - The budget for this project is £180, and 30 unit credits worth of time, equating to 240 hours of time commitment.

Both of these factors will mean that the final solution favours simpler, more complete solutions. Anything novel is a potential failure point that could mean the project does not complete.

2.3 Survey of existing tracking technologies

In order to fully evaluate the myriad of arm tracking methods available, a Pugh Matrix has been created (see Table 1) to evaluate and compare several factors between each method. Each method will score $1 - 5$ in each category, with a brief comment justifying the score in each column.

Below are the decision factors of the Pugh Matrix and a brief description of how they are scored:

Complexity - This is calculated based on a well reasoned estimate of the total time to build the system. Each multiple of 48 hours reduces the score by 1, e.g. $0 - 48$ hours would score 5, $49 - 96$ hours would score 4.

Cheapness - This is calculated based on a well reasoned estimate of the total cost of building the system. Each multiple of £36 reduces the score by 1, e.g. £$0 -$ £36 would score 5, £$37 -$ £72 would score 4.

Accuracy - This is calculated from available research on the accuracy of this method. Each multiple of $5°$ above $5°$ reduces the score by 1, e.g. an absolute accuracy of $6 - 10°$ would score 4, $11 - 15°$ would score 3.

Latency - This is calculated from available research on the time it takes to obtain a single measurement of all 7 joint angles from this method. Each multiple of 100 ms above 300 ms reduces the score by 1, e.g. a latency of 301 ms $- 400$ ms would score 4.

Analysis of existing methods shows that the two computer vision approaches score similarly in terms of viability, therefore further analysis will be required to confirm the viability of each method. These will be performed in the form of accuracy and latency experiments. This has the advantage of being able to examine more factors than existing literature - of particular interest is the accuracy of measuring multiple joints angles simultaneously.

In particular, the potential accuracy advantage of the marker tracking method makes it more appealing, though this is offset by the stupendous cost of conventional Vicon camera systems. However, open source software along with ArUco markers could be used to offset this cost - which is worth exploring before committing to any substantial development. For example, **ref:arucomarkers** describes using ArUco markers along with standard home cameras to perform specialised kinematic analysis **ref:arucomarkers**.

Table 1. Pugh Matrix with detailed scoring justifications

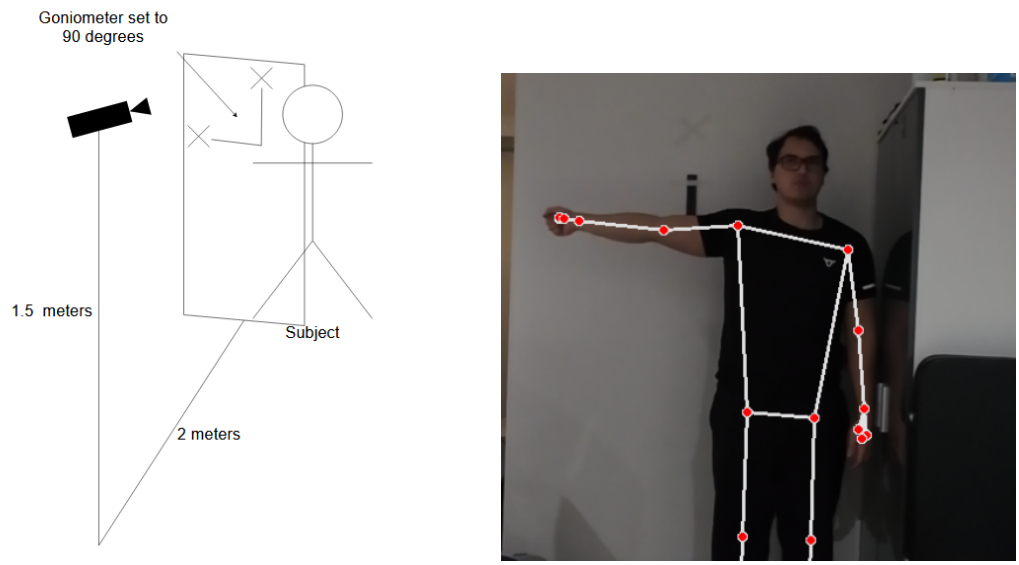
Method	**Complexity** $1-5$	**Cheapness** $1-5$	**Accuracy** $1-5$	**Latency** $(1-5)$	**Total**
Pose estimation via computer vision	5 Existing solutions reduce complexity $30-40$ hours	5 £30 USB Camera	3 $10^\circ - 15^\circ$ **ref:cverror**	5 ≤ 100 ms **ref:cvlatency**	18
Marker tracking via computer vision	5 Very complex, though can be simplified through FOSS	3 £180+ for Vicon, cheaper solutions exist	5 <5° error	5 100-200ms	18
IMU-based	3 100 hours including PCB design + drift correction	4 £$50 -$ £60	3 2.8° **ref:imuerror**	5 <1 ms **ref:imudatasheet**	15
Flex Sensors	2 150 hours	4 £40-60	4 10° **ref:bendaccuracy**	5 ≤ 100 ms	15

3 Comparing two viable computer vision based methods of robot arm teleoperation

The criteria selected both ArUco marker tracking and pose estimation as equally viable methods of human arm tracking. This section aims to develop a minimum viable product (MVP) version of both methods to be used to quantify which approach is more suitable for the final solution.

3.1 Comparison of both computer vision methods

In order to evaluate the accuracy of both computer vision methods, a simple experiment was set up to measure the absolute accuracy on the joint angles that can be measured. A goniometer is attached to the wall at $90°$, with two end positions marked at $90°$ and $180°$. The subject will move their arm repeatedly between these markers, and the recording is captured by a camera as described in Fig. 2.



(a) A diagram of the experimental setup. **(b)** Example data capture

Fig. 2. The experimental setup, showing data capture.

The subject will then flex and extend their elbow between the two endpoints. In order to obtain a sufficient amount of data, 5 videos are recorded, each showing 10 flexion and extensions.

This creates enough samples to perform ANOVA between each video, to maintain experimental integrity. In order to measure error, the graph of elbow flexion angles is extended towards the $90°$

and $180°$ endpoints. This secondary graph is then subtracted from the first graph to obtain the error value, from which we can calculate the root mean square error, and is shown in Fig. 3.

Fig. 3. Showing how RMSE is calculated from elbow flexion data.

Upon analysis of the RMSE, it was found that this value was $25.83°$. which was greater than the literature value of $10.37°$ **ref:cverror**. Frame by frame analysis of the video showed that this descrepancy was not attributable to human error - the arm was clearly flexing and extending to the correct endpoints.

Plotting the inferred joint positions frame by frame revealed the culprit - perspective! The machine learning model used for joint position extraction is finding extra movement in the depth axis of the video, and attributing it to roll in the elbow axis, instead of flexion. This is clearly shown in Fig. 4.

To analyse the performance of joint angle tracking in the three dimensional case, a simple movement path is created. A 7 DoF manipulator is created from relevant DH parameters (specifics found in appendix Table 3) These movements are then fed through a forward kinematics processing program to obtain a visualisation of this movement path, and this movement is recreated by the subject.

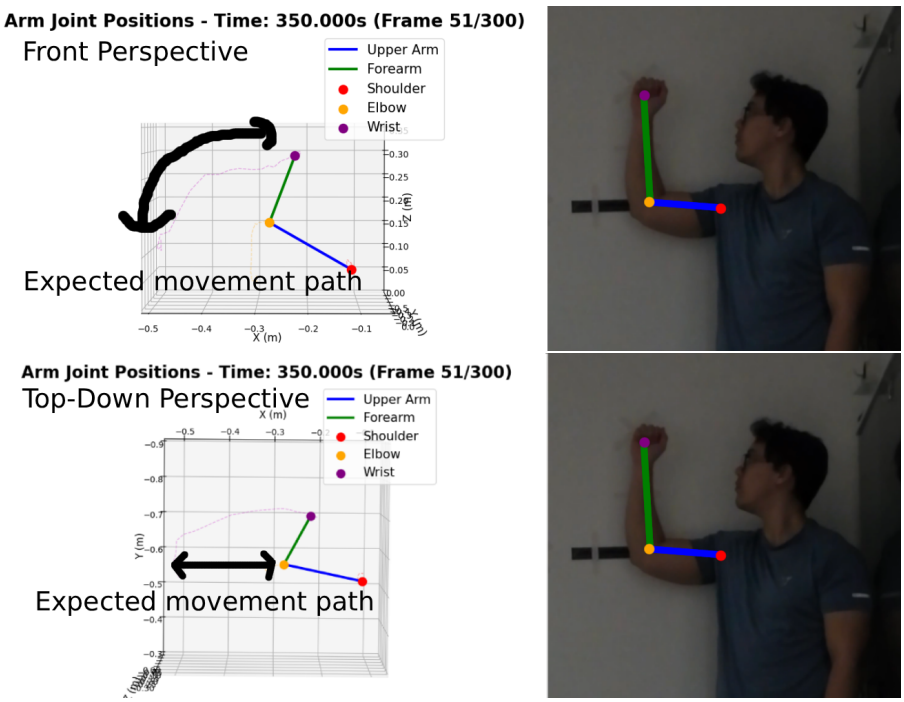


Fig. 4. Front and top view of inferred 3D joint positions.

This path is shown in 5. This movement has been selected to be easily recreated, whilst showing movement in three different joint angles simultaneously. The same data processing for the two dimensional elbow flex case is used to obtain RMSE of three different joint angle movements.

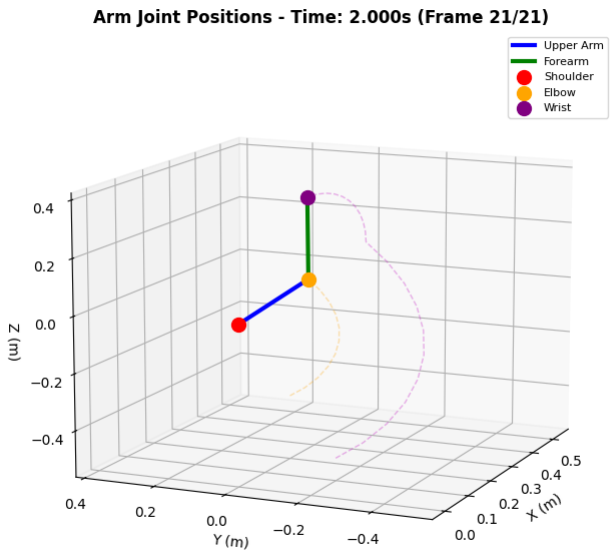


Fig. 5. 3D Joint Motion Path

3.2 Results of joint angle tracking methods

Table 2 shows the compiled RMSE errors for tracking each joint in both the two dimension elbow flexing case and the three dimension movement path.

Table 2. RMSE error of either tracking method.

Method	**RMSE Error Elbow Flexion 2D (°)**	**RMSE Error Elbow Flexion 3D (°)**	**RMSE Error Shoulder Flexion (°)**	**RMSE Error Shoulder Abduction (°)**
Pose estimation	25.83	7.19	6.89	11.68
Marker Tracking	N/A	N/A	N/A	N/A

Whilst the two dimensional RMSE error is larger than the literature suggests, the three dimensional joint error shows better accuracy than expected. Apart from the shoulder adduction/abduction measurements, each joint being tracked is almost in the same order of magnitude of the $3°$ requirements of the project.

Despite various attempts, ArUco marker tracking seemed to be too unstable in this specific application. Both 2D paper markers and 3D printed markers were used, though the glossy surface finish of either meant that tracking was unreliable at close distances and impossible at moderate distances. This was obtained from an experiment wherein a marker is held up to the camera and rotated around various axes, measuring the captured rotation.

Fig. 6 shows a plot of the measured roll angle, and shows the gaps in which no measurement was made. Likewise, the background conditions of the image also made a major difference to whether or not the marker was tracked. For example, Fig. 7 shows the marker not tracking whilst not distorted by angle at all, compared to being tracked at a very large angle. Clearly this level of unreliability is unsuitable for the final tracking task, and therefore no further testing will be required.

An interesting aspect that was encountered when performing the accuracy characterisation experiments with pose estimation is shown in Fig. 8, and points to a downside not immediately apparent. The program uses machine learning to create an estimation of joint locations in an image and was ostensibly trained on a large corpus of human images. Hence, there was trouble distinguishing between real humans and pictures of humans being worn on the subject's clothing - a downside in the favour of the marker tracking method.

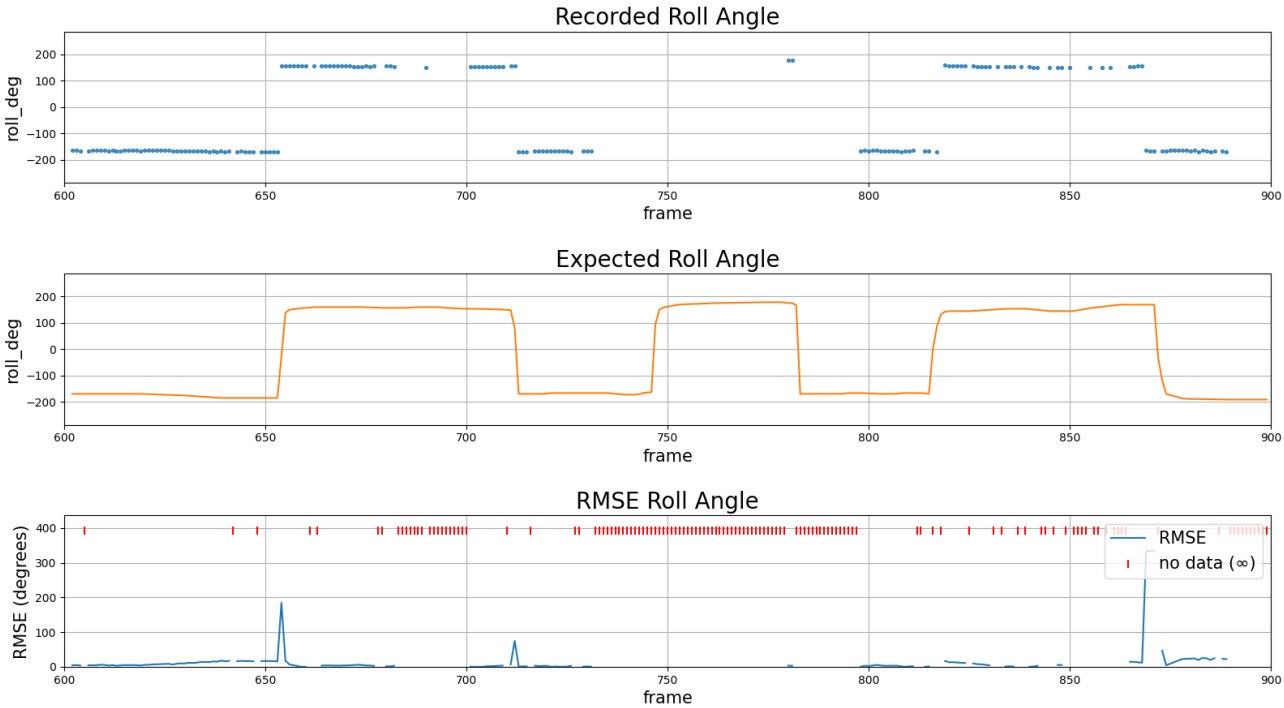


Fig. 6. Measured vs Expected graph of roll angle of ArUco marker.

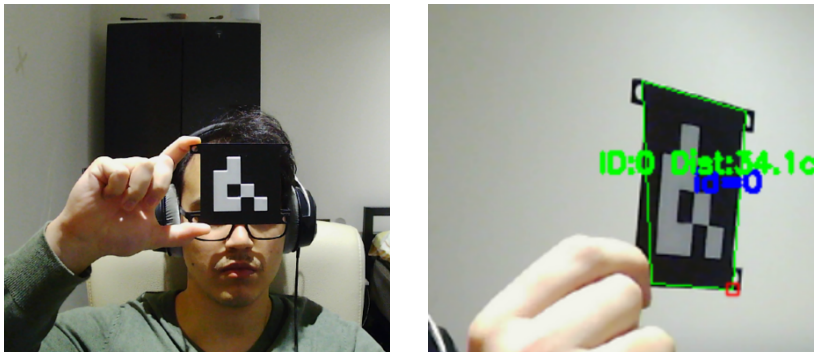


Fig. 7. Background dependence of the ArUco marker.

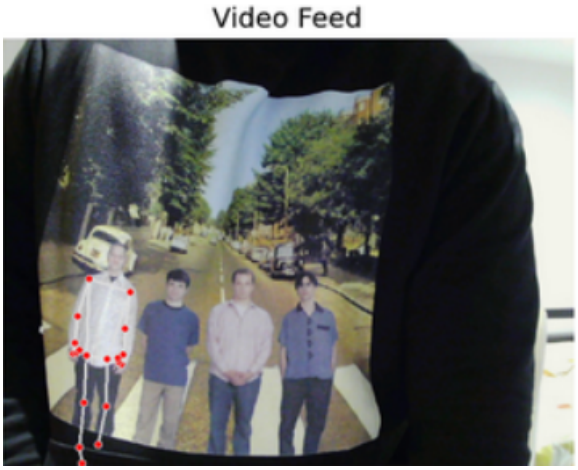


Fig. 8. Pose estimation software mistaking a picture of humans printed on clothing for an actual person.

3.3 Final Tracking Method

Results show that the pose estimation model is not able to achieve the 3° required for the project, it is able to get within the same order of magnitude of accuracy. Moreover, the mediapipe model

used to power this is optimised for real time measurement; using a 30 fps camera, each measurement takes 33 ms, well below the required 300 ms. Hence, the comparison between conventional joystick control and human arm tracking teleoperation will use the pose estimation approach for comparison.

4 Arm tracking vs conventional control experiments

This section explains three scenarios that have been devised in CoppeliaSim to try and characterise the performance of each tracking method.

4.1 Description of experiment to compare conventional vs human arm tracking

In order to compare fairly the two methods of control, three scenarios will be examined:

Section 4.2 - Position matching task similar to **ref:movementrange ref:movementrange**. A random location within the envelope of the arm is selected and the operator must control the arm to this location as fast as possible.

Section 4.3 - Pick and place task. An object and target location spawns within the envelope of the arm, and the operator must manipulate the object to the target.

Section 4.4 - Obstacle laden pick and place task. Similar to pick and place task, but with tracking of obstacle collisions.

The operator must stand further away from the screen to be in frame of the camera for the pose estimation method. To negate the perception advantages this would incur for the controller tasks, the controller method will be performed sitting this same distance away.

To measure the performance of the position matching task, the distance covered is divided by the time taken to find the normalised speed measurement.

4.2 Scenario 1: Random position matching task

In this scenario a random position within the reachable envelope of the arm is marked and the operator must move the grippers of the arm to within 0.06m of this location as fast as they can. This

time is then recorded, and then used to divide the distance where the arm started from to find the average speed of the arm during the experiment.

100 samples were taken, with 10 runs of 10 experiments. Analysis of variance (ANOVA) shows no significant deviation between each run (see appendix for detailed ANOVA results). The controller method was roughly 42% faster than the pose estimation algorithm, although with a lower standard deviation of 0.043 compared to 0.053.

The speed difference is likely attributable to the operator's familiarity of using controllers for other tasks. Notably, the difference in standard deviation could imply that pose estimation is stronger for tasks that require consistency, rather than outright speed. However, this standard deviation figure is small enough to be attributable to noise, if nothing else.

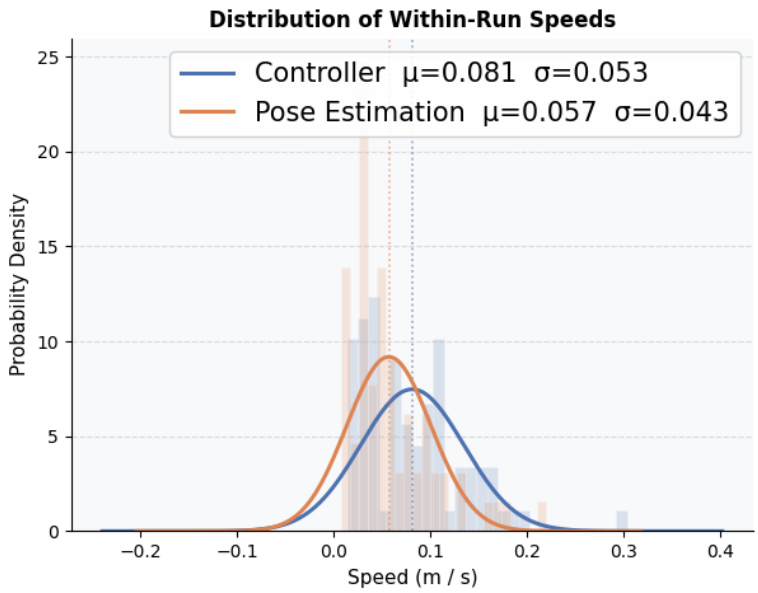


Fig. 9. Standard distribution plot of the results of either experiment.

4.3 Scenario 2: Simple environment pick and place task

Scenarios 2 and 3 consist of four different phases shown in Fig. 10 The object is considered 'held' as long as YuMi's grippers are within a certain distance threshold, and transition from an open state to a closed state. This prevents the experiment from being limited by the design and simulation of a suitable end effector - though this may act as an interesting extension for future research.

To measure the performance of the pick and place task, the cumulative time spent in each phase is measured, along with the overall time to complete the task. Likewise, to ensure fairness between

each method in both pick and place tasks, seeded randomness will be used to ensure that both methods get the same trial spawn position.

The pose estimation algorithm shows 60.3% reduction in time over the controller control in this task, with the noticeable difference being in the proportion of time spent in the 'approach' phase (see Fig. 11) The controller method requires the operator to orient the gripper over the object, and doing this in certain arm positions can lead to singularities in the inverse kinematics solver, hence the longer time.

These singularities can be seen in a joint angle over time plot of one of the experiments in Fig. 12. The advantage of the pose estimation method is that the human arm of the operator acts as a safety net for unreachable positions.

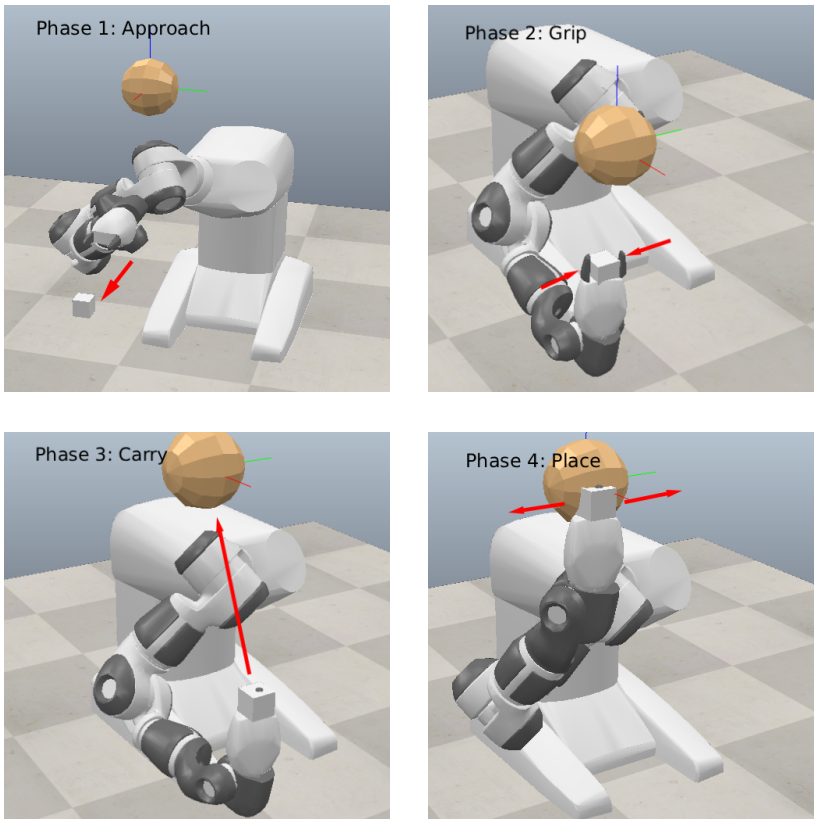


Fig. 10. Phases of the pick and place scenario

4.4 Scenario 3: Obstacle-rich environment pick and place task

The pose estimation algorithm is quicker on average by 57.2% in this task, likely for the same reasons as above - where the controller solution is running into singularities. Additionally, the added dexterity shown led to 3.46 fewer collisions (4.14 vs 7.60) on average compared to the controller

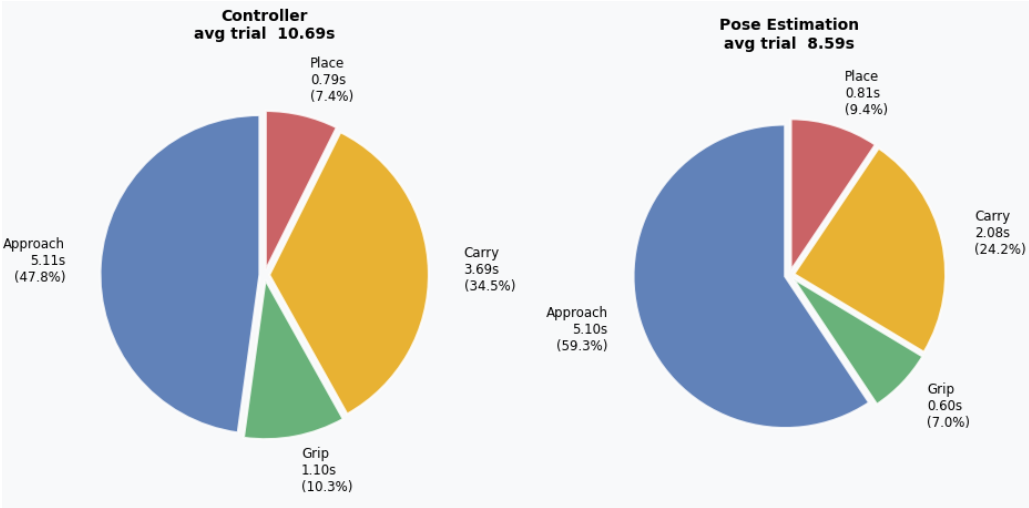


Fig. 11. Average time spent in each phase for either control method.

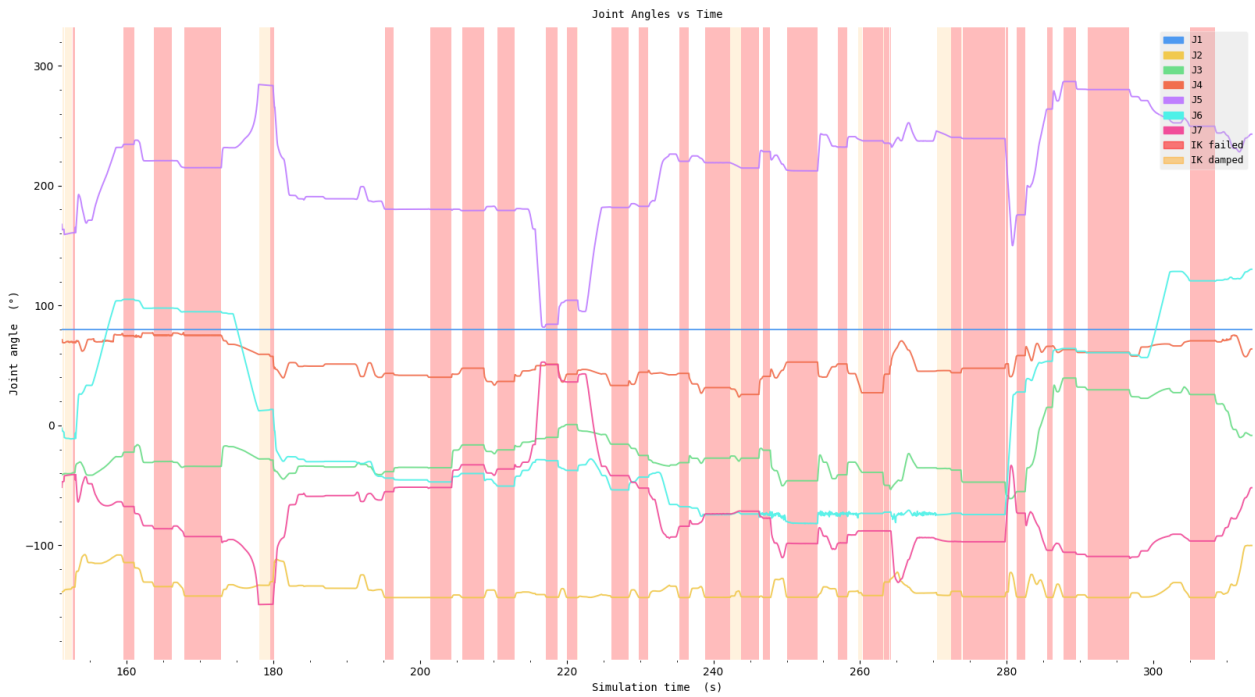


Fig. 12. Joint angle over time plot of the robot arm during a controller experiment

solution.

The fewer collisions may be due to more control that the operator has over the entire arm, as more data can be transmitted from the operator to the robot, in a way that is intuitive. Likewise, the smooth control of the pose estimation algorithm may make it more predictable compared to the IK approach of the controller solution - where it may be jumping over singularity positions unpredictably.

The proportion of time spent in the approach phase increased by 9% for the pose estimation solution, and decreased by 66.8% for the controller solution. Interestingly, the controller solution becomes more balanced in the time spent in each movement phase, whereas the pose estimation

solution becomes front loaded.

There is also a noticeable difference in the average trial time between scenario 3 and scenario 2, likely attributable to the operator becoming more skilled and experience with each control method and experiment.

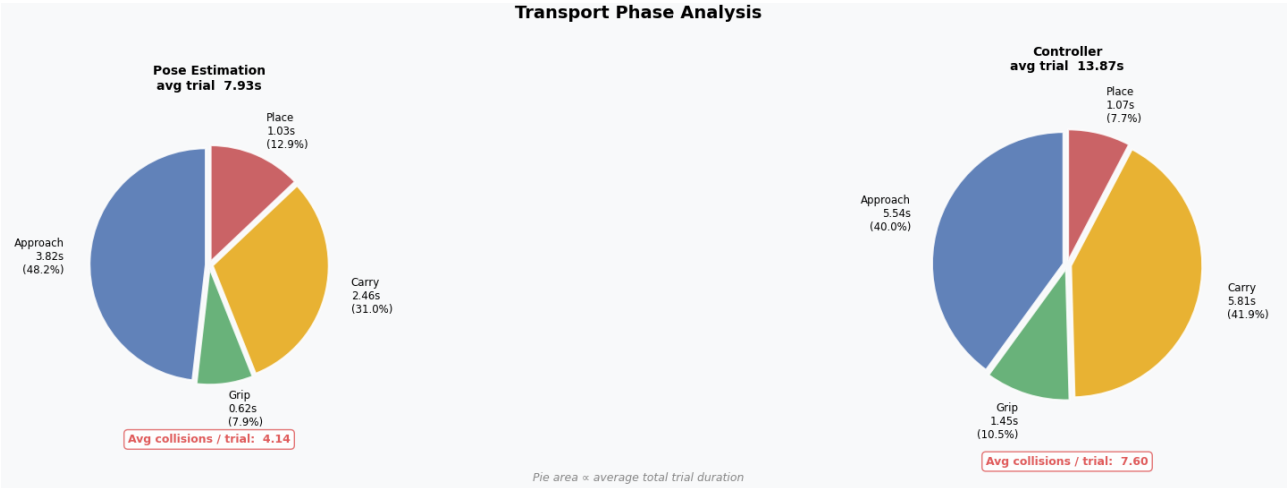


Fig. 13. Standard distribution plot of the results of either experiment

5 Conclusions and future work

5.1 Conclusions

5.2 Future work

Appendices

A Project outline

A.1 Overview and motivations

The goal of this project is to create a system for controlling a laterally mounted robot arm with a human arm. Ideally, the system will map each degree of freedom (DoF) of the human arm to the robot arm.

The motivation for this project are the two books: Revitalizing Human-Robot Interaction (Zhou et al) and Human Robot Awareness and Robot Teleoperation (Honghao et al), which describes research into remote teleoperation of laterally mounted robot arms. These books explain the use of an inertial measurement unit (IMU) device which maps the movement of human arms to robot arms, however they only map the end effector position to the position of the wrist.

My project would aim to create a device allowing for mapping of the full robot arm to an operator's arm. In an obstacle-rich environment, mapping of the end effector position is insufficient. The main advantage of teleoperation is to allow the human operator to make real time judgments for complex situations, and thus fully mapping out the operator's intentions to the robot is crucial.

One such scenario that is researched is given in the paper by Jin et al, describing a complex obstacle avoidance algorithm for moving a robot arm underwater for sample retrieval. The drawback mentioned in this paper is that both the sample and arm origin is stationary, and the algorithm takes up to ten seconds to solve this simple case. Ostensibly, one can imagine the complexity required for a scenario where both the sample, arm, and obstacles are moving - solving this autonomously is inefficient compared to using human intuition.

A.2 Deliverables

The deliverables for this project are as follows:

- The creation of a device or software that is capable of reading the angles in an operator's arm, in real time.

- Part of the project scope will be the rigorous examination of existing methods, and justifying the chosen method.
- Applying the values read in from this software to a simulation environment, where a laterally mounted 7 DoF robot arm is controlled from the operator's arm movements.
- Creating an obstacle-rich simulation, where an operator is tasked with moving between points to quantify the differences in success criteria over conventional control methods (e.g. a joystick or traditional end effector mapping).
- Creation of a final report that outlines a table of contents, introduction, methodology, results and conclusion.

B Risk assessment

MANCHESTER 1824
The University of Manchester

RISK REGISTER FOR 3rd YEAR PROJECT

Project Title:	Human Arm Tracking Human Robot Interface		Submission Date:	17/10/25	
Student Name:	Nicholas	Teague			

Project Risk	Severity			Potential			Score (Severity x Potential)	Mitigation Measures
	L	M	H	L	M	H	L=1, M=2, H=3	
Repetitive Strain Injuries from robot operation		x			x		4	Limit testing sessions to less than 30 minutes, ensuring that the operator is adequately warmed up with stretches. Pay attention to muscle discomfort and fatigue, immediately stopping all activity if experienced.
Muscle strains from attempted movements beyond joint capabilities		x			x		4	Program software to show operator limits for joint positions, design control mapping to stay comfortably within human movement ranges.
Eye strain and postural fatigue			x		x		6	Take regular breaks from seated work, e.g. walking around the room, focusing eyes on a far away point, etc.
Mental exhaustion		x				x	6	Schedule regular breaks from work, and avoid working at obscene hours in the day. Vary work tasks, and try to maintain a sufficient work-life balance and sleep schedule.
Motion sickness		x		x			2	Minimise system latency, immediately stop tests if discomfort occurs. Avoid extended testing situations.
Collision with objects in workspace	x					x	3	Keep work area, especially areas in the workspace of the arm clutter-free.

C Code

All relevant code (and source for this report) can be found on my GitHub @ https://github.com/coding-beagle/UoMFinalProject.git

While at it, please check out https://www.nteague.com

D DH Parameters

The DH parameters used for the arm are as follows:

Table 3. Denavit-Hartenberg Parameters for 7-DoF Arm (where L_u is upper arm length and L_l is lower arm length)

Joint	Description	θ_i	α_i	d_i	a_i
1	Shoulder abduction/adduction	q_1	$\frac{\pi}{2}$	0	0
2	Shoulder flexion/extension	q_2	$\frac{\pi}{2}$	0	0
3	Shoulder rotation	q_3	$-\frac{\pi}{2}$	0	L_u
4	Elbow flexion/extension	q_4	$\frac{\pi}{2}$	0	0
5	Forearm rotation	q_5	$-\frac{\pi}{2}$	0	L_l
6	Wrist flexion/extension	q_6	$\frac{\pi}{2}$	0	0
7	Wrist radial/ulnar deviation	q_7	0	0	0

E ANOVA Figures